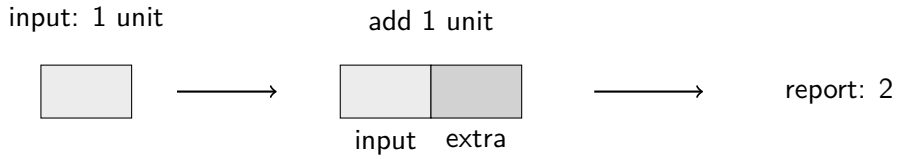
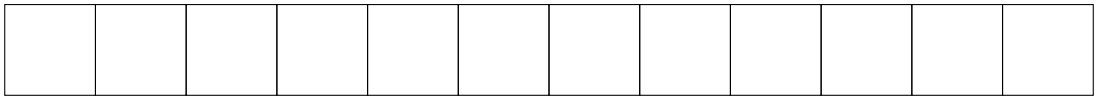
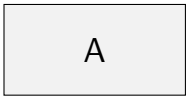
A machine reports a strip's length plus a secret extra number: 1, 2, 3, 4, 5, or 6. The extra number stays the same for the whole round. All lengths are whole units. Each square in the paper row is one unit.



Problem 1: Play the machine. Secretly choose an extra number. Your partner gives you a strip; tell them its report. They may ask for more strips until they can name your extra number. Reveal it and switch jobs.

Round	Strips requested	Reports
1		
2		
3		
4		











Problem 2: Two hidden strips give these reports. Use unit cubes to try possible lengths, one strip at a time. Find every possible pair of lengths.

P reports 7	Q reports 10

Length of P	Length of Q	Extra number

Problem 3: Could P be longer than Q? What can you know about the two lengths without learning the extra number?

Problem 4: Each row uses a different machine. A known strip gives report 8. Two hidden strips give reports 9 and 12. Build the hidden strips for each row.

Known strip	Length that reports 9	Length that reports 12
4 units $\longrightarrow$ 8		
6 units $\longrightarrow$ 8		
2 units $\longrightarrow$ 8		

Problem 5: Your partner knows only that two hidden strips report 7 and 10. Secretly choose a machine and write both hidden lengths under a cover. Let your partner request the report for one known strip, then build each hidden strip. Reveal the lengths and switch jobs.

Problem 6: Each box claims to come from one machine. Which boxes could be right? Repair each impossible box by changing just one report.

I	II	III
2 units $\to$ 5	2 units $\to$ 8	2 units $\to$ 3
4 units $\to$ 7	4 units $\to$ 10	4 units $\to$ 6
6 units $\to$ 10	6 units $\to$ 12	6 units $\to$ 7

Problem 7: Make three length-and-report pairs with exactly one wrong report. Give it to your partner to repair. The correct machine must use one of the six extra numbers.